Introduction to Hadoop

The evolution of data processing takes the route of DBMS → RDBMS → DATA WARE HOUSE → DATA LAKE. One of the important aspects of this development is to incorporate flexibility in processing different kinds of data.

As we defined Big Data in terms of volume, high velocity and extensible variety of data. The data in it will be of three types:

1) Structured Data, 2) Semi Structured Data 3) Un-structured Data.

Structured Data:

It concerns all data which can be stored in a database and are addressable using ~~stu~~ structured Query Language (SQL). They can be arranged in a table and there exists relational key among the data stored in different rows and columns.

Let us take an example:

To select all columns from a table (Customer) for rows where the ~~Last Name~~ Last-Name columns has Smith for its value, one should send a select statement to the server back end:

SELECT * FROM customers WHERE LAST-Name='Smith'.

The server back end would reply with a result set -

+-------+--------+--------+-------+

| Cust_No | Last_Name | First_Name |

+-----+------+-----+-----+

1001	Smith	John
2039	Smith	David
2098	Smith	Mathew

+-----+------+-----+-----+

3 rows in set (0.05 sec)

This search is possible since all the last name are stored in a column and the corresponding first name are stored in another column and there is a relational key between the two columns. This type of data are called the structured data.

structured data is often managed using structured Query Language (SQL).

Characteristics of structured data:

1). structured data first depends on creating a data model - a model of the types of business data that will be recorded and how they will be stored, processed and accessed.

2) It needs to specify what fields of data will be stored and how that data will be stored:

3. Data type like numeric, currency, alphabetic, date address etc.

4. It should also needs to specify any restrictions on the data input.

5. Structured data has the advantage of being easily entered, stored, queried and analyzed.

6. SQL has been a standard of the American National Committee for Information Technology Standards (INCITS) Technical Committee DM 32 — Data Management and Interchange.

Semi structured Data:

Semi-structured data is information that does not reside in a relational database that does have some organizational proper-ties that makes it easier to analyse.

Ex: XML, JSON documents, e-mail etc. NoSQL databases are considered as semi structure.

Unstructured Data:

Unstructured Data refers to information that either does not have a predefined data model or is not organized in a pre-defined manner. Unstructured data is typically text-heavy, but may contain data such as dates, numbers as well. This results in irregularities and ambiguities that make it difficult to understand using traditional programs as compared to data stored in fielded form in data base or annotated in documents.

Examples of unstructured data are:
Books, Documents, metadata, health records, audio, video, analog ~~and~~ data images etc.

The processing of such unstructured data along with structured and semi struc-tured data is efficiently done in the frame work of Hadoop.

Big Data and Hadoop:

Apache Hadoop is an open-source software frame work written in Java for distributed storage and distributed processing of very large data sets on computer clusters.

Hadoop allows to store and process big data in a distributed environment across clusters of computers using simple programming methods.

it is designed to scale up from single servers to thousands of machines, each offering local computation and storage.

* ~~##~~ The Hadoop architecture is fault tallarent in the sence, that it ~~assu~~ is designed with a fundamental assumption that hardware failures are common and should be automatically handled by the frame work.

For many years, users who want to store and analyze data would store the data in a database and process it via SQL ~~service~~. queries. The Web has changed most of the assumptions of this era. On the ~~wo~~ Web the data is unstructured and large, and the database can neither capture the data into a schema nor scale it to store and process it.

HADOOP ARCHITECTURE

The core of Apache Hadoop consists of a storage part, known as Hadoop Distributed File system (HDFS) and a processing part called MapReduce. Hadoop split files into large blocks and distribute them accross nodes in a cluster. In each node the data are processed parallely. The basic Apache Hadoop framework is composed of the following modules:

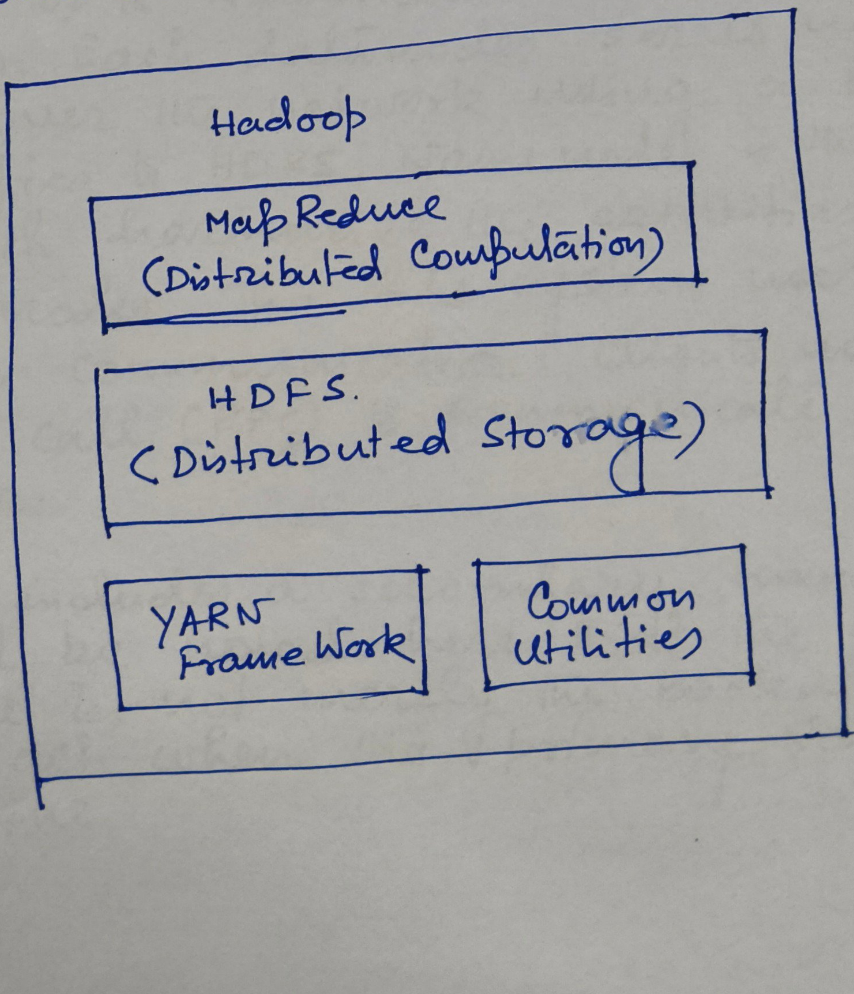
1. Hadoop Distributed File System (HDFS): It is a distributed file-system that stores data on commodity machines, providing very high aggregate bandwidth accress the clusters.

it is to be noted here that if the nodes are geographically located at a particular location ~~them is~~ and ~~o~~ posses common configuration, they are called as cluster. On the other hand if they ~~er~~ remain separated geographically, they are called the grid.

2. Hadoop YARN — A resource-management platform responsible for managing computing resources in clusters and using them for scheduling of users applications; ~~and~~

3. Hadoop MapReduce — A method used for distri-buted processing.

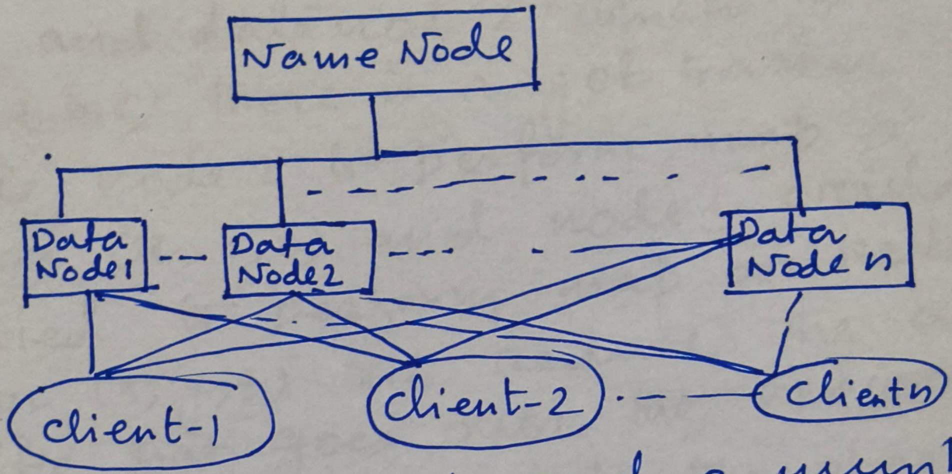
The schematic diagram of Hadoop echosystem is as follows:



Hadoop Distributed File System (HDFS)

The Hadoop distributed file system (HDFS) is a distributed, scalable and portable file system written in java for the Hadoop framework.

The simplistic view of Hadoop cluster is as follows



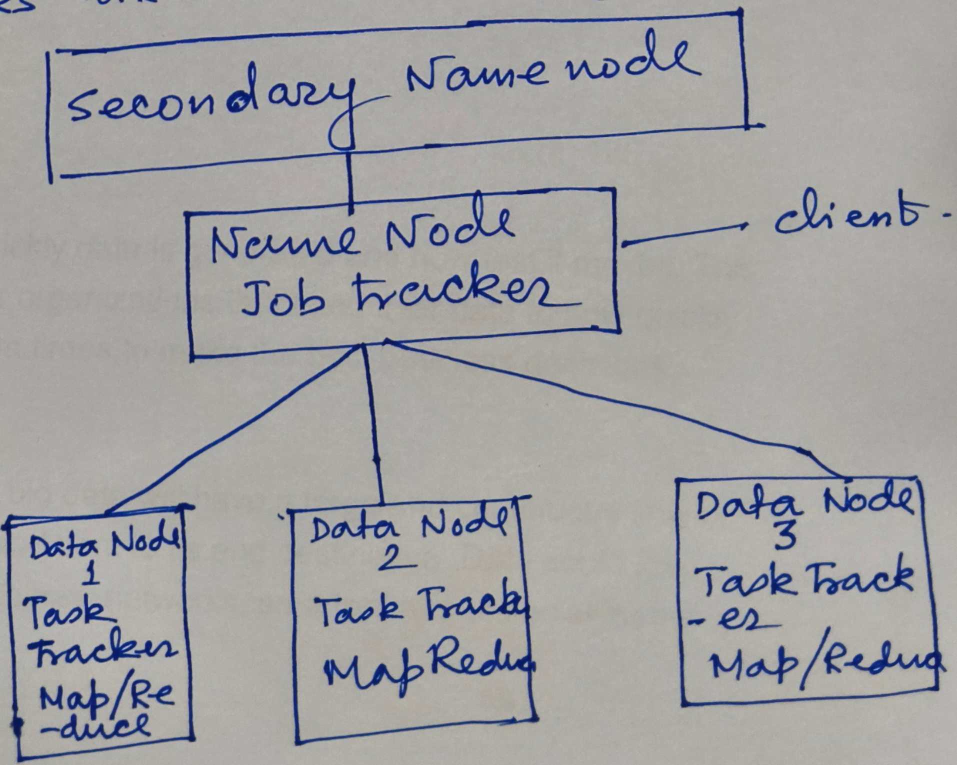
It consists of a Namenode and a number of data nodes. Each datanodes serves up blocks of data over the network using a block protocol specifies to HDFS. Name node is the master node which harmonize the activities of all the datanodes. The file system uses TCP/IP sockets for communication. Clients uses remote procedure call (RPC) to communicate among each other.

The HDFS includes a secondary namenode. It should be noted here that the secondary Namenode is not merely the backup name node to act when the primary Namenode goes offline.

The secondary namenode regularly connects with the primary namenodes and builds snapshots of the directory information of the primary name node. It is utilized by the system to manage the local and remote the directories.

suppose the data node 1 contains data (files) (x, y, z) and data node 2 which contains data (a, b, c) There is a job tracker which scedules node 2 to perform map or reduce tasks on (a, b, c) and node 1 would be scheduled to perform map or reduce task on (x, y, z). This reduces the amount of traffic that goes over the network and prevents unnecessary data transfer.

Finally the architecture of Apache Hadoop HDFS looks like the following:



What are the 5 V's of big data?

The 5 V's of big data -- velocity, volume, value, variety and veracity -- are the five main and innate characteristics of big data. Knowing the 5 V's lets data scientists derive more value from their data while also allowing their organizations to become more customer-centric.

Earlier this century, big data was talked about in terms of the three V's -- volume, velocity and variety. Over time, two more V's -- value and veracity -- were added to help data scientists more effectively articulate and communicate the important characteristics of big data. In some cases, there's even a sixth V term for big data -- variability.

The 5 V's are defined as follows:

1. *Velocity* is the speed at which the data is created and how fast it moves.
2. *Volume* is the amount of data qualifying as big data.
3. *Value* is the value the data provides.
4. *Variety* is the diversity that exists in the types of data.
5. *Veracity* is the data's quality and accuracy.

Velocity

Velocity refers to how quickly data is generated and how fast it moves. This is an important aspect for organizations that need their data to flow quickly, so it's available at the right times to make the best business decisions possible.

An organization that uses big data will have a large and continuous flow of data that's being created and sent to its end destination. Data could flow from sources such as machines, networks, smartphones or social media.

Velocity applies to the speed at which this information arrives -- for example, how many social media posts per day are ingested -- as well as the speed at which it needs to be digested and analyzed -- often quickly and sometimes in near real time.

As an example, in healthcare, many medical devices today are designed to monitor patients and collect data. From in-hospital medical equipment to wearable devices, collected data needs to be sent to its destination and analyzed quickly.

In some cases, however, it might be better to have a limited set of collected data than to collect more data than an organization can handle -- because this can lead to slower data velocities.

Volume

Volume refers to the amount of data that exists. Volume is like the base of big data, as it's the initial size and amount of data that's collected. If the volume of data is large enough, it can be considered big data. However, what's considered to be big data is relative and will change depending on the available computing power that's on the market.

For example, a company that operates hundreds of stores across several states generates millions of transactions per day. This qualifies as big data, and the average number of total transactions per day across stores represents its volume.

Value

Value refers to the benefits that big data can provide, and it relates directly to what organizations can do with that collected data. Being able to pull value from big data is a requirement, as the value of big data increases significantly depending on the insights that can be gained from it.

Organizations can use big data tools to gather and analyze the data, but how they derive value from that data should be unique to them. Tools like

Apache Hadoop can help organizations store, clean and rapidly process this massive amount of data.

A great example of big data value can be found in the gathering of individual customer data. When a company can profile its customers, it can personalize their experience in marketing and sales, improving the efficiency of contacts and garnering greater customer satisfaction.

Variety

Variety refers to the diversity of data types. An organization might obtain data from several data sources, which might vary in value. Data can come from sources in and outside an enterprise as well. The challenge in variety concerns the standardization and distribution of all data being collected.

As noted above, collected data can be unstructured, semi-structured or structured. Unstructured data is data that's unorganized and comes in different files or formats. Typically, unstructured data isn't a good fit for a mainstream relational database because it doesn't fit into conventional data models. Semi-structured data is data that hasn't been organized into a specialized repository but has associated information, such as metadata. This makes it easier to process than unstructured data. Structured data, meanwhile, is data that has been organized into a formatted repository. This means the data is made more addressable for effective data processing and analysis.

Raw data also qualifies as a data type. While raw data can fall into other categories -- structured, semi-structured or unstructured -- it's considered *raw* if it has received no processing at all. Most often, raw applies to data imported from other organizations or submitted or entered by users. Social media data often falls into this category.

A more specific example could be found in a company that gathers a variety of data about its customers. This can include structured data culled from transactions or unstructured social media posts and call center text. Much of this might arrive in the form of raw data, requiring cleaning before processing.

Veracity

Veracity refers to the quality, accuracy, integrity and credibility of data. Gathered data could have missing pieces, might be inaccurate or might not be able to provide real, valuable insight. Veracity, overall, refers to the level of trust there is in the collected data.

Data can sometimes become messy and difficult to use. A large amount of data can cause more confusion than insights if it's incomplete. For example, in the medical field, if data about what drugs a patient is taking is incomplete, the patient's life could be endangered.

Both value and veracity help define the quality and insights gathered from data. Thresholds for the *truth* of data often -- and should -- exist in an organization at the executive level, to determine whether it's suitable for high-level decision-making.

Where might a red flag appear over the veracity data? It could, for instance, be lacking in proper *data lineage* -- that is, a verifiable trace of its origins and movement.